

Occurrence of Livestock-Associated Methicillin-Resistant Staphylococcus aureus in Turkey and Broiler Barns and Contamination of Air and Soil Surfaces in Their Vicinity

The emission of microorganisms, especially resistant bacteria such as methicillin-resistant *Staphylococcus aureus* (MRSA), from poultry farms is of public interest, and its occurrence and relevance are controversially discussed. So far, there are limited data on this issue. In this study, we investigated the occurrence of livestock-associated (LA)-MRSA inside and outside previously tested MRSA-positive poultry barns in Germany. In total, five turkey and two broiler fattening farms were investigated four and three times, respectively. In a longitudinal study during one fattening period, samples were collected from animals, the animals' environment inside the barn, including the air, and the barns' surroundings, such as ambient air and boot swabs of ground surfaces at different distances from the barn. Moreover, a cross-sectional study was carried out once inside the barns on five turkey and four broiler farms during the last third of the fattening period. In the cross-sectional study, LA-MRSA was detected in the air of most barns (7 of 9, 77.8%), as well as in many samples originating from animals, with detections levels of 50 to 54% in broiler and 62 to 77% in turkey farms. In the longitudinal study, LA-MRSA was found in the ambient air outside two turkey barns and on the ground surface on the downwind side of many (44.4%) turkey and broiler farms. The same spa types of isolates were observed inside and outside the barns. Transmission of MRSA within poultry farms, as well as emission via the airborne route, seems to be possible.